

(c) angle FBC ,

Answer ° [2]

(d) angle DFO .

Answer ° [2]

- 15 The table below shows the mass of 500 packets of 1 kg chocolate packed by machine A.

Mass (x grams)	$980 < x \leq 990$	$990 < x \leq 1000$	$1000 < x \leq 1010$	$1010 < x \leq 1020$
Frequency	74	178	182	66

- (a) Calculate an estimate of the mean mass.

Answer g [1]

- (b) Calculate an estimate of the standard deviation.

Answer g [1]

- (c) Explain why the mean and standard deviation are estimates.

Answer

[1]

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- (d) The mean and standard deviation of mass of chocolate packed by machine B is given below.

Mean (grams)	1000.2
Standard Deviation (grams)	12.5

If you are the owner of the chocolate distribution business, which machine will you buy? Explain your reason.

Answer

[1]

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- 16 (a) Express 150 as a product of its prime factors.

Answer [1]

- (b) Find the smallest integer a , such that $(150+150^2)a$ is a perfect square.

Answer $a =$ [2]